

=====

=====

ROOF/OS — MASTER PLAYBOOK

ROADMAP + EXECUTION

PURPOSE

This document is divided into two distinct operating layers:

ROADMAP

= Where ROOF/OS is going.

EXECUTION

= How ROOF/OS gets there safely and verifiably.

The roadmap must not be confused with the current sprint.

A phase can exist on the roadmap without being active.

Execution determines what is being worked on NOW.

=====

=====

PART I — ROADMAP

1. MISSION

Build ROOF/OS into a dependable roofing operating system connecting:

LEAD

→ INSPECTION

- PHOTOS
- AI OBSERVATION
- TECHNICIAN VERIFICATION
- MEASUREMENT
- ESTIMATE
- REPORT
- PRODUCTION
- WARRANTY

2. ARCHITECTURE INVARIANT

AI observes.

ROOF/OS validates.

Geometry calculates.

Technician verifies.

Estimator prices.

AI IS NEVER AUTHORITATIVE FOR:

- ☐ Measurements
- ☐ Geometry
- ☐ Pricing
- ☐ Insurance decisions
- ☐ Engineering determinations
- ☐ Code-compliance determinations

3. BUSINESS TARGET

NATIONAL CUSTOMER TARGET:

50 states

×

10 customers per state

500 customers

ANNUAL PRICE:

\$500 per customer per year

ANNUAL RECURRING REVENUE TARGET:

500 customers

×

\$500/year

\$250,000/year

BUSINESS TARGET:

500 paying customers

at

\$500/year

\$250,000 ARR

UNIT ECONOMICS TARGET:

1 customer

\$500/year

10 customers

\$5,000/year

100 customers

\$50,000/year

500 customers

\$250,000/year

GEOGRAPHIC TARGET:

10 customers per state
across
50 states

IMPORTANT:

The \$250,000 target is based on 500 customers, not 50 customers.

The state calculation is:

$50 \times 10 = 500$ customers.

The revenue calculation is:

$500 \times \$500 = \$250,000/\text{year}$.

4. PHASE ROADMAP — 0 THROUGH 47

PHASE 0 — KICKOFF / RECOVERY

Establish a known, recoverable starting state.

- ☐ Canonical repository identified
- ☐ Branch identified
- ☐ Commit identified
- ☐ Worktree verified
- ☐ Backup created
- ☐ Remote backup verified
- ☐ Baseline recorded

PHASE 1 — REPOSITORY AUDIT

Understand what actually exists.

- ☐ Application structure
- ☐ API routes
- ☐ Database migrations
- ☐ Authentication
- ☐ Storage
- ☐ Security
- ☐ Tests
- ☐ Mobile
- ☐ Deployment
- ☐ Documentation
- ☐ Known gaps

PHASE 2 — ARCHITECTURE AUDIT

Verify system boundaries and data ownership.

- ☐ AI boundary
- ☐ Measurement boundary
- ☐ Pricing boundary
- ☐ Insurance boundary
- ☐ Technician authority
- ☐ Workspace isolation
- ☐ API boundaries

PHASE 3 — DATABASE / RLS

Establish data integrity and isolation.

- ☐ Schema
- ☐ Migration order
- ☐ Workspace isolation
- ☐ RLS
- ☐ Storage policies
- ☐ Foreign keys
- ☐ Constraints
- ☐ Indexes
- ☐ Audit fields

PHASE 4 — SECURITY BASELINE

Remove unsafe trust boundaries.

- ☐ Authentication
- ☐ Authorization

- ☐ Workspace membership
- ☐ Input validation
- ☐ Request limits
- ☐ Timeout controls
- ☐ Secret handling
- ☐ Error sanitization
- ☐ CSP
- ☐ Storage access

PHASE 5 — CORE APPLICATION FOUNDATION

Make the base application reliable.

- ☐ Navigation
- ☐ Workspace handling
- ☐ API conventions
- ☐ Error handling
- ☐ Loading states
- ☐ Data access patterns

PHASE 6 — LEAD / CRM

Create the operational customer pipeline.

- ☐ Lead creation
- ☐ Customer records
- ☐ Contact information
- ☐ Lead status
- ☐ Assignment
- ☐ Notes
- ☐ Follow-up
- ☐ Workspace isolation

PHASE 7 — INSPECTION FOUNDATION

Create the field inspection record.

- ☐ Inspection creation
- ☐ Property association
- ☐ Technician association
- ☐ Inspection status
- ☐ Inspection photos
- ☐ Metadata
- ☐ Inspection history

PHASE 8 — PHOTO SYSTEM

Make inspection photography reliable.

- ☐ Upload
- ☐ Storage
- ☐ MIME validation
- ☐ Size limits
- ☐ Metadata
- ☐ Workspace isolation
- ☐ Inspection association
- ☐ Retrieval
- ☐ Deletion rules

PHASE 9 — D.2 AI VISION CONTRACT

Define exactly what AI can and cannot say.

- ☐ Contract
- ☐ Strict schema
- ☐ Enumerations
- ☐ Forbidden fields
- ☐ Forbidden authority claims
- ☐ Photo references
- ☐ Damage references
- ☐ Validation
- ☐ Contract tests

PHASE 10 — D.3 PHOTO AI PIPELINE

Safely execute AI photo analysis.

- ☐ Authentication
- ☐ Workspace authorization
- ☐ Workflow authorization
- ☐ Photo authorization
- ☐ Image limits
- ☐ Canonical hash
- ☐ Cache
- ☐ Gemini request
- ☐ Timeout
- ☐ Response parsing
- ☐ D.2 validation
- ☐ Persistence
- ☐ Error handling

PHASE 11 — AUTHENTICATION

Restore reliable production login.

- ☐ Magic link
- ☐ OTP fallback where implemented
- ☐ Callback handling
- ☐ Safe next handling
- ☐ Rate-limit handling
- ☐ Error handling
- ☐ Production redirect configuration
- ☐ Production login test

PHASE 12 — PROPERTY / GIS INTELLIGENCE

Connect property context to the workflow.

- ☐ Address normalization
- ☐ Geocoding
- ☐ Property lookup
- ☐ Building footprint
- ☐ Map/GIS context
- ☐ Weather/storm context
- ☐ Provenance

PHASE 13 — MEASUREMENT SYSTEM

Create authoritative measurement workflow.

- ☐ Technician measurements
- ☐ Eave dimensions
- ☐ Rafter information
- ☐ Pitch
- ☐ Geometry inputs
- ☐ Waste
- ☐ Squares
- ☐ Gutters
- ☐ Soffit
- ☐ Fascia

AI remains non-authoritative.

PHASE 14 — TECHNICIAN VERIFICATION

Create the human authority gate.

- ☐ Review AI observations
- ☐ Verify damage

- ☐ Verify pitch
- ☐ Verify measurements
- ☐ Correct observations
- ☐ Record technician
- ☐ Record verification time
- ☐ Lock authoritative values

PHASE 15 — GEOMETRY ENGINE

Turn verified field data into mathematical measurements.

- ☐ Facet geometry
- ☐ Pitch calculations
- ☐ Area calculations
- ☐ Waste calculations
- ☐ Eave/rake/ridge calculations
- ☐ Gutter calculations
- ☐ Soffit/fascia calculations
- ☐ Validation

PHASE 16 — ESTIMATING ENGINE

Generate estimates from verified data.

- ☐ Material quantities
- ☐ Labor
- ☐ Accessories
- ☐ Waste
- ☐ Disposal
- ☐ Overhead
- ☐ Margin
- ☐ Estimate totals
- ☐ Calculation audit trail

PHASE 17 — PRICING SYSTEM

Make pricing controlled and explainable.

- ☐ Owner price book
- ☐ Material pricing
- ☐ Labor pricing
- ☐ Vendor pricing
- ☐ Pricing timestamps
- ☐ Margin controls
- ☐ Regional adjustments
- ☐ Pricing audit

PHASE 18 — CLAIMS / INSURANCE WORKFLOW

Support claim-related workflow without allowing AI to make authoritative insurance decisions.

- ☐ Claim information
- ☐ Documents
- ☐ Damage evidence
- ☐ Scope comparison
- ☐ Supplement workflow
- ☐ Human review
- ☐ Audit trail

PHASE 19 — REPORTING

Generate useful customer and operational reports.

- ☐ Inspection report
- ☐ Photo report
- ☐ Estimate report
- ☐ Technician report
- ☐ Claim support report
- ☐ PDF/output
- ☐ Report history

PHASE 20 — SUPPLEMENTS

Support legitimate scope-change workflows.

- ☐ Original scope
- ☐ New observations
- ☐ Verified changes
- ☐ Documentation
- ☐ Supplement calculation
- ☐ Approval workflow
- ☐ Audit trail

PHASE 21 — PRODUCTION WORKFLOW

Connect sold work to execution.

- ☐ Approved estimate
- ☐ Job creation
- ☐ Scheduling
- ☐ Crew assignment
- ☐ Materials

- ☐ Production status
- ☐ Completion
- ☐ Closeout

PHASE 22 — WARRANTY

Track post-installation responsibility.

- ☐ Warranty record
- ☐ Installation details
- ☐ Photos
- ☐ Materials
- ☐ Completion date
- ☐ Customer information
- ☐ Warranty events

PHASE 23 — MOBILE FIELD APPLICATION

Create dependable field operation.

- ☐ Mobile authentication
- ☐ Offline data
- ☐ Offline photos
- ☐ Local queue
- ☐ Sync
- ☐ Conflict handling
- ☐ Retry
- ☐ Field measurements
- ☐ Technician verification

PHASE 24 — OFFLINE-FIRST HARDENING

Ensure field operations survive poor connectivity.

- ☐ Offline inspection
- ☐ Offline photo capture
- ☐ Offline measurement
- ☐ Offline notes
- ☐ Sync queue
- ☐ Retry
- ☐ Duplicate prevention
- ☐ Conflict resolution
- ☐ Recovery after interruption

PHASE 25 — API / INTEGRATION HARDENING

Make external integrations safe and reliable.

- ☐ Timeouts
- ☐ Retries
- ☐ Rate limits
- ☐ Validation
- ☐ Error handling
- ☐ Secret management
- ☐ Provider failure behavior
- ☐ Provenance

PHASE 26 — SECURITY HARDENING

Attack the application before attackers do.

- ☐ Authorization tests
- ☐ Workspace isolation
- ☐ Storage tests
- ☐ Input fuzzing
- ☐ Secret exposure review
- ☐ Error leakage review
- ☐ Rate-limit review
- ☐ Dependency audit
- ☐ CSP review

PHASE 27 — PERFORMANCE

Make common workflows production-capable.

- ☐ API latency
- ☐ Database queries
- ☐ Image processing
- ☐ AI timeout behavior
- ☐ Cache effectiveness
- ☐ Mobile performance
- ☐ Bundle size

PHASE 28 — OBSERVABILITY

Know what the system is doing.

- ☐ Structured logging
- ☐ Error tracking
- ☐ Request IDs
- ☐ AI analysis IDs
- ☐ Workflow history

- ☐ Audit events
- ☐ Health checks
- ☐ Operational dashboard

PHASE 29 — TESTING SYSTEM

Create repeatable evidence.

- ☐ Unit tests
- ☐ Contract tests
- ☐ API tests
- ☐ Security tests
- ☐ Database tests
- ☐ Mobile tests
- ☐ Offline tests
- ☐ End-to-end tests
- ☐ Production smoke tests

PHASE 30 — END-TO-END WORKFLOW

Prove:

LEAD

- INSPECTION
- PHOTOS
- AI
- TECHNICIAN
- MEASUREMENT
- GEOMETRY
- ESTIMATE
- REPORT
- JOB
- CLOSEOUT

PHASE 31 — DEPLOYMENT

Make production deployment repeatable.

- ☐ Build
- ☐ Environment variables
- ☐ Database migrations
- ☐ Deployment
- ☐ Domain
- ☐ Authentication redirects
- ☐ Smoke tests
- ☐ Rollback plan

PHASE 32 — PRODUCTION VERIFICATION

Prove the deployed system works.

- ☐ Production HTTP
- ☐ Login
- ☐ Workspace
- ☐ Lead
- ☐ Inspection
- ☐ Photo upload
- ☐ AI workflow
- ☐ Technician verification
- ☐ Measurement
- ☐ Estimate
- ☐ Report
- ☐ Mobile

PHASE 33 — PILOT

Run real roofing workflows.

- ☐ Real lead
- ☐ Real property
- ☐ Real inspection
- ☐ Real photos
- ☐ Real technician
- ☐ Real measurement
- ☐ Real estimate
- ☐ Real report
- ☐ Capture failures

PHASE 34 — PILOT HARDENING

Convert field failures into engineering work.

OBSERVE

- RECORD
- REPRODUCE
- FIX
- TEST
- COMMIT
- DEPLOY
- VERIFY

PHASE 35 — CUSTOMER ONBOARDING

Create repeatable customer adoption.

- ☐ Account setup
- ☐ Workspace setup
- ☐ Users
- ☐ Roles
- ☐ Training
- ☐ First inspection
- ☐ First estimate
- ☐ First report

PHASE 36 — COMMERCIALIZATION

Turn the product into a repeatable business.

- ☐ Pricing
- ☐ Sales material
- ☐ Demo
- ☐ Onboarding
- ☐ Support
- ☐ Billing
- ☐ Customer success
- ☐ Renewal

PHASE 37 — AI IMPROVEMENT LOOP

Improve AI without weakening authority boundaries.

FIELD DATA

- REVIEW
- ERROR CLASSIFICATION
- CONTRACT-SAFE IMPROVEMENT
- TEST
- DEPLOY
- MONITOR

PHASE 38 — AUTOMATION

Automate repetitive operational work.

- ☐ Lead routing
- ☐ Notifications
- ☐ Report generation
- ☐ Follow-up
- ☐ Data synchronization

- ☐ Monitoring
- ☐ Operational alerts

PHASE 39 — AGENT / AUTOMATION LAYER

Agents may:

- ☐ Observe
- ☐ Organize
- ☐ Summarize
- ☐ Route
- ☐ Recommend
- ☐ Trigger approved workflows

Agents must not bypass:

- ☐ Authorization
- ☐ Human verification
- ☐ Pricing controls
- ☐ Measurement authority
- ☐ Security controls

PHASE 40 — SCALE

Prepare for increased customers and workloads.

- ☐ Database scaling
- ☐ Storage scaling
- ☐ AI scaling
- ☐ Queueing
- ☐ Caching
- ☐ Monitoring
- ☐ Cost controls

PHASE 41 — MULTI-TENANT HARDENING

Ensure one roofing company cannot access another company's data.

- ☐ Workspace isolation
- ☐ RLS
- ☐ Storage isolation
- ☐ API authorization
- ☐ Background jobs
- ☐ Reporting
- ☐ AI cache isolation

PHASE 42 — NATIONAL EXPANSION

Support operation across states.

- ☐ Regional pricing
- ☐ Regional codes
- ☐ Regional workflows
- ☐ Weather data
- ☐ Vendor differences
- ☐ Tax/business requirements
- ☐ Customer onboarding
- ☐ Support

PHASE 43 — OPERATING SYSTEM MATURITY

Move from product to operating platform.

- ☐ Standard operating procedures
- ☐ Automated health checks
- ☐ Customer success
- ☐ Operational analytics
- ☐ Revenue analytics
- ☐ Reliability targets
- ☐ Disaster recovery

PHASE 44 — SCALE / PLAYOFFS

Stress the system.

- ☐ High customer volume
- ☐ High photo volume
- ☐ AI failures
- ☐ Database failures
- ☐ Storage failures
- ☐ Provider outages
- ☐ Authentication failures
- ☐ Network failures
- ☐ Mobile offline scenarios

PHASE 45 — FINAL HARDENING

Close remaining critical gaps.

- ☐ Security
- ☐ Reliability
- ☐ Recovery

- ☐ Data integrity
- ☐ AI boundaries
- ☐ Production monitoring
- ☐ Documentation
- ☐ Support

PHASE 46 — FINAL PLAY

Prepare ROOF/OS for large-scale operation.

- ☐ Production verified
- ☐ Recovery verified
- ☐ Customer onboarding repeatable
- ☐ Field workflow repeatable
- ☐ AI workflow verified
- ☐ Pricing verified
- ☐ Reporting verified
- ☐ Monitoring verified
- ☐ Business process verified

PHASE 47 — SUPER BOWL

ROOF/OS operates as a dependable roofing operating system.

SUCCESS REQUIRES:

- ☐ Real customers
- ☐ Real field usage
- ☐ Real revenue
- ☐ Stable production
- ☐ Verified security
- ☐ Verified AI boundaries
- ☐ Reliable measurements
- ☐ Controlled estimating
- ☐ Repeatable onboarding
- ☐ Disaster recovery
- ☐ National scalability
- ☐ Evidence-driven improvement

=====

=====

PART II — EXECUTION

5. EXECUTIVE OPERATING RULE

The roadmap describes the destination.

Execution describes the CURRENT WORK.

Only the active milestone should consume primary engineering attention.

DO NOT WORK PHASE 20 WHILE PHASE 10 HAS AN UNRESOLVED CRITICAL BLOCKER UNLESS THE DEPENDENCY IS EXPLICITLY DOCUMENTED.

6. PERMANENT DEVELOPMENT RULES

RULE 1 — BACKUP FIRST

Before substantial code changes:

BACKUP

→ VERIFY BACKUP EXISTS

→ THEN CODE

RULE 2 — NO LOST WORK

Every meaningful completed change must exist in a durable, externally recoverable location.

Preferred:

Git commit + remote branch.

RULE 3 — VERIFY, DO NOT ASSUME

Never claim:

[] Backed up

[] Pushed

[] Deployed

[] Tested

[] Fixed

[] Production-ready

[] Verified

unless actual evidence exists.

RULE 4 — SOURCE OF TRUTH

Use actual:

- [] Repository state
- [] Git state
- [] Configuration
- [] Logs
- [] Test output
- [] Database results
- [] Production responses

Do not invent missing facts.

RULE 5 — NO BLIND CODING

AUDIT

- PLAN
- CHANGE
- TEST
- VERIFY

RULE 6 — ONE CHANGE SET AT A TIME

Every meaningful milestone has:

OWNER

+

TIME BOX

+

DEFINITION OF DONE

+

EVIDENCE

RULE 7 — FAILURES ARE RECORDED

Failures are not hidden.

Blockers are assigned.

Unknown is not converted to PASS.

RULE 8 — PRODUCTION IS A SEPARATE GATE

Passing local tests does not equal production verification.

RULE 9 — AI IS NEVER AUTHORITATIVE

AI output must remain inside the D.2 contract.

RULE 10 — RECOVERABILITY BEFORE VELOCITY

Fast work that cannot be recovered is unacceptable.

=====

=====

7. BACKUP / RECOVERY PROTOCOL

MANDATORY SEQUENCE:

1. Inspect repository.
2. Identify current branch.
3. Identify current commit.
4. Confirm worktree state.
5. Create backup branch or equivalent.
6. Push backup.
7. Verify remote SHA.
8. Record evidence.
9. Only then begin substantial changes.

CHECKLIST:

- ☐ Repository identified
- ☐ Correct remote verified
- ☐ Current branch known
- ☐ Current commit known
- ☐ Worktree state known
- ☐ Backup branch created
- ☐ Backup pushed
- ☐ Remote SHA verified
- ☐ Recovery path documented

IF REMOTE BACKUP CANNOT BE VERIFIED:

STOP BEFORE SUBSTANTIAL CODE CHANGES.

=====

=====

8. CURRENT EXECUTION PRIORITY

CURRENT PRIORITY ORDER:

1. Protect the codebase.
2. Verify the recovered state.
3. Finish and verify D.3 photo AI.
4. Restore and verify production authentication.
5. Eliminate critical security and production blockers.
6. Prove the complete roofing workflow.
7. Deploy and verify production.
8. Run real-world pilot jobs.
9. Harden based on evidence.
10. Scale.

=====

=====

9. FIRST-WEEK EXECUTION PLAN

DAY 1 — RECOVERY + BASELINE

OWNER:

Engineering lead

OBJECTIVE:

Establish a known, recoverable repository state.

- ☐ Confirm canonical repo
- ☐ Confirm branch
- ☐ Confirm commit
- ☐ Confirm worktree
- ☐ Create backup branch

- ☐ Push backup
- ☐ Verify remote SHA
- ☐ Record baseline
- ☐ Audit current state

DONE WHEN:

Remote backup is independently verified and repository state is documented.

DAY 2 — SECURITY + ARCHITECTURE

OBJECTIVE:

Verify boundaries before adding functionality.

- ☐ Authentication
- ☐ Workspace authorization
- ☐ Storage
- ☐ RLS
- ☐ API security
- ☐ AI authority boundary
- ☐ Existing tests

DONE WHEN:

Critical trust boundaries are understood and blockers are assigned.

DAY 3 — D.2 + D.3

OBJECTIVE:

Finish the controlled AI vision layer.

- ☐ D.2 contract
- ☐ D.2 validator
- ☐ D.2 tests
- ☐ D.3 endpoint
- ☐ Photo authorization
- ☐ Limits
- ☐ Hashing
- ☐ Cache
- ☐ Gemini request
- ☐ Timeout
- ☐ Persistence
- ☐ Security tests

DONE WHEN:

D.2 tests pass and D.3 passes all available automated checks.

DAY 4 — AUTH + CORE WORKFLOW

OBJECTIVE:

Restore the user path.

- ☐ Login
- ☐ Callback
- ☐ OTP/magic-link behavior
- ☐ Workspace
- ☐ Lead
- ☐ Inspection
- ☐ Photos

DONE WHEN:

An authenticated user can enter the application and begin a roofing workflow.

DAY 5 — MEASUREMENT + ESTIMATE

OBJECTIVE:

Prove the authority chain.

- ☐ Technician verification
- ☐ Measurements
- ☐ Geometry
- ☐ Estimate
- ☐ Pricing
- ☐ Audit trail

DONE WHEN:

Verified field data can produce an auditable estimate without AI becoming the authority.

DAY 6 — END-TO-END TEST

OBJECTIVE:

Run:

LEAD

- INSPECTION
- PHOTOS
- AI
- TECHNICIAN
- MEASUREMENT
- ESTIMATE
- REPORT

DONE WHEN:

Objective test evidence exists and every failure is recorded.

DAY 7 — PRODUCTION / PILOT GATE

OBJECTIVE:

Determine what is actually ready.

- ☐ Production deployment
- ☐ Production login
- ☐ Production workflow
- ☐ Security smoke test
- ☐ AI status
- ☐ Database status
- ☐ Mobile status
- ☐ Remaining blockers
- ☐ Next milestone

DONE WHEN:

Every unresolved issue has an owner, priority, and next action.

=====

=====

10. D.2 — AI VISION CONTRACT

PURPOSE

D.2 defines exactly what ROOF/OS allows AI vision to observe, how observations are structured, and what AI is forbidden from claiming.

D.2 IS THE CONTRACT.

D.2 DOES NOT MAKE AI AUTHORITATIVE.

INPUT

One or more authorized inspection photographs.

OUTPUT

A validated RoofAIObservationPacket containing non-authoritative visual observations.

D.2 MAY CONTAIN:

- [] Contract version
- [] Analysis timestamp
- [] Model identifier
- [] Measurement status
- [] Authority disclaimer
- [] Image audit
- [] Roof classification
- [] Facet observations
- [] Edge observations
- [] Penetration observations
- [] Damage observations
- [] Summary
- [] Warnings

=====

=====

11. D.2 — CONCRETE SCHEMA EXAMPLE

EXAMPLE / ILLUSTRATIVE OUTPUT:

```
{
  "contract_version": "1.0.0",
  "analysis": {
    "model": "gemini-2.5-flash",
    "analyzed_at": "2026-09-30T12:00:00Z",
    "measurement_status": "unverified_ai_observation"
  },
  "disclaimer": "NON-AUTHORITATIVE VISUAL OBSERVATIONS ONLY. Not certified measurements, engineering determinations, insurance damage verifications, or pricing calculations. All observations require independent physical verification by a licensed roofing technician.",
  "photos": [
    {
      "photo_id": "photo_01",
      "usable": true,
      "quality": "good",
      "roof_visible": true,
      "notes": "Primary front roof plane is visible."
    },
    {
      "photo_id": "photo_02",
      "usable": true,
      "quality": "fair",
      "roof_visible": true,
      "notes": "Right-side roof plane partially visible."
    }
  ],
  "roof": {
    "style": "gable",
    "primary_material": "asphalt_shingle",
    "confidence": "high"
  },
  "facets": [
    {
      "facet_id": "facet_01",
      "label": "front_primary",
      "orientation": "front",
      "pitch_class": "moderate",
      "pitch_observation": "appears moderate",
      "confidence": "medium",
      "source_photo_ids": ["photo_01"]
    },
    {
      "facet_id": "facet_02",
```

```
"label": "right_plane",
"orientation": "right",
"pitch_class": "moderate",
"pitch_observation": "appears moderate",
"confidence": "medium",
"source_photo_ids": ["photo_02"]
},
],
"edges": [
{
"edge_type": "eave",
"visible": true,
"source_photo_ids": ["photo_01"]
},
{
"edge_type": "rake",
"visible": true,
"source_photo_ids": ["photo_01"]
},
{
"edge_type": "ridge",
"visible": true,
"source_photo_ids": ["photo_01"]
}
],
"penetrations": [
{
"penetration_id": "penetration_01",
"type": "plumbing_vent",
"confidence": "high",
"source_photo_ids": ["photo_01"]
}
],
"damage": [
{
"damage_id": "damage_01",
"type": "missing_shingle",
"confidence": "medium",
"source_photo_ids": ["photo_01"],
"notes": "Visually apparent missing shingle area."
}
],
"summary": {
"observations": [
```

```

    "Asphalt-shingle gable roof is visually apparent.",
    "One plumbing vent is visible.",
    "A possible missing shingle is visible on the front plane."
  ],
  "warnings": [
    "Pitch and damage observations require technician verification."
  ]
}
}

```

NOTE:

This is an illustrative schema example. The exact production contract must be verified against the implementation before this example is treated as authoritative.

```

=====

=====

```

12. D.2 — FORBIDDEN OUTPUT

AI MUST NOT PROVIDE AUTHORITATIVE OR COMMERCIAL VALUES FOR:

```

[ ] roof_squares
[ ] squares
[ ] roof_area_sqft
[ ] facet_area
[ ] area_sqft
[ ] area
[ ] length_ft
[ ] linear_feet
[ ] quantity
[ ] price
[ ] cost
[ ] estimate_amount
[ ] total_price
[ ] insurance_coverage
[ ] claim_decision
[ ] code_compliance
[ ] certified_measurement

```

- ☐ certified_pitch
- ☐ payout
- ☐ deductible
- ☐ acv
- ☐ rcv
- ☐ depreciation

AI MUST NOT PRESENT OBSERVATIONS AS:

- ☐ Certified
- ☐ Official
- ☐ Guaranteed
- ☐ Engineered
- ☐ Exact
- ☐ Authoritative

=====

=====

13. D.2 — VALIDATION

Every AI packet must pass:

- ☐ Exact contract version
- ☐ Required-field validation
- ☐ Allowed-field validation
- ☐ Data-type validation
- ☐ Enum validation
- ☐ ID validation
- ☐ Photo-reference validation
- ☐ Facet-reference validation
- ☐ Timestamp validation
- ☐ Recursive forbidden-key scan
- ☐ Authority-language checks
- ☐ Damage source-photo requirements

D.2 DEFINITION OF DONE:

- ☐ Contract implemented
- ☐ Concrete schema validated
- ☐ Prohibited fields rejected

- [] Authority claims rejected
- [] Photo references validated
- [] Contract tests pass
- [] Commit created
- [] Commit pushed
- [] Remote commit verified

=====

=====

14. D.3 — PHOTO AI ANALYSIS PIPELINE

PURPOSE

D.3 turns the D.2 contract into the actual server-side photo-analysis workflow.

D.2 defines WHAT AI is allowed to say.

D.3 defines HOW ROOF/OS safely obtains, validates, caches, stores, and returns that observation.

EXECUTION FLOW:

AUTHORIZED USER

↓

WORKSPACE AUTHORIZATION

↓

WORKFLOW AUTHORIZATION

↓

AUTHORIZED PHOTO LOOKUP

↓

AUTHORIZED PHOTO RETRIEVAL

↓

IMAGE LIMIT CHECKS

↓

CANONICAL CONTENT HASH

↓

CACHE CHECK

↓

GEMINI VISION ANALYSIS

↓

PARSE PROVIDER RESPONSE

↓

D.2 CONTRACT VALIDATION

↓

SAFE PERSISTENCE

↓

RETURN VALIDATED OBSERVATION

D.3 AUTHENTICATION:

☐ User authenticated

☐ Workspace exists

☐ User belongs to workspace

☐ Requested workflow belongs to workspace

D.3 PHOTO AUTHORIZATION:

☐ Photo belongs to workspace

☐ Photo is authorized inspection photo

☐ Supported storage bucket

☐ Supported MIME type

☐ Valid metadata

☐ Authorized for requested workflow

D.3 LIMITS:

MAX PHOTOS:

50

MAX CUMULATIVE IMAGE DATA:

30 MB

SUPPORTED:

☐ JPEG

☐ PNG

☐ WebP

D.3 CANONICAL HASH:

SHA-256 raw bytes

↓

lowercase hashes

↓

sort hashes

↓

join with ":"

↓

SHA-256 UTF-8 joined string

Hash must not depend on:

- ☐ Paths
- ☐ Filenames
- ☐ IDs
- ☐ Timestamps
- ☐ Original ordering

D.3 CACHE:

Cache may be used only when:

- ☐ forceRefresh = false
- ☐ Canonical hash matches
- ☐ Stored analysis exists
- ☐ Stored analysis passes D.2 validation

Invalid cached analysis MUST NOT be trusted.

D.3 FORCE REFRESH:

forceRefresh = true

bypasses the existing AI cache.

D.3 PROVIDER:

Gemini 2.5 Flash

Provider credentials remain server-side.

D.3 PROVIDER SAFETY:

- ☐ Non-authoritative output
- ☐ No roof squares
- ☐ No roof area
- ☐ No linear feet
- ☐ No quantities
- ☐ No pricing
- ☐ No insurance decisions
- ☐ No code compliance
- ☐ Apparent pitch is visual only

- ☐ Damage requires source photographs
- ☐ Unknown remains unknown
- ☐ No invented observations

D.3 TIMEOUT:

30 seconds target.

D.3 RESPONSE ACCEPTANCE:

- ☐ Parse successfully
- ☐ Intended observation packet
- ☐ D.2 validation
- ☐ Forbidden-key scan
- ☐ Type validation
- ☐ Enum validation
- ☐ Photo-reference validation

HTTP success from the AI provider is NOT sufficient.

D.2 validation is the acceptance gate.

D.3 PERSISTENCE:

- ☐ ai_analysis
- ☐ ai_analyzed_at
- ☐ ai_model_version
- ☐ ai_content_hash
- ☐ updated_at

D.3 MUST NOT silently overwrite technician-authoritative fields.

D.3 DEFINITION OF DONE:

- ☐ Authentication works
- ☐ Workspace authorization works
- ☐ Workflow authorization works
- ☐ Photo authorization works
- ☐ Storage access controlled
- ☐ 50-photo limit enforced
- ☐ 30 MB limit enforced
- ☐ MIME restrictions enforced
- ☐ Canonical hashing works
- ☐ Cache works
- ☐ Force refresh works
- ☐ Provider timeout enforced

- [] Provider output passes D.2
- [] Only intended AI fields persist
- [] Errors sanitized
- [] Typecheck passes
- [] Endpoint tests pass
- [] Security tests pass
- [] Commit created
- [] Commit pushed
- [] Remote commit verified

=====

=====

15. D.2 VS D.3

D.2 = CONTRACT

Defines what AI may observe and what ROOF/OS will accept.

D.3 = PIPELINE

Obtains photos, calls AI, validates against D.2, caches, persists, and returns the result.

D.2 = RULES.

D.3 = EXECUTION.

=====

=====

16. AI AUTHORITY BOUNDARY

PHOTO

↓

D.3

↓
AI PROVIDER
↓
D.2 VALIDATOR
↓
NON-AUTHORITATIVE AI OBSERVATION
↓
TECHNICIAN REVIEW
↓
VERIFIED FIELD DATA
↓
GEOMETRY
↓
ESTIMATING
↓
PRICE

AI OBSERVATION
≠
VERIFIED MEASUREMENT

AI OBSERVATION
≠
PRICE

AI OBSERVATION
≠
INSURANCE DECISION

AI OBSERVATION
≠
ENGINEERING DETERMINATION

=====

=====

17. EVIDENCE STANDARD

A claim is not a completion.

ACCEPTABLE EVIDENCE:

- ☐ Command output
- ☐ Test result
- ☐ Git commit
- ☐ Remote Git verification
- ☐ Database result
- ☐ Production response
- ☐ Browser verification
- ☐ Deployment evidence
- ☐ Real workflow transaction

NO EVIDENCE:

NOT VERIFIED.

=====

=====

18. TESTING GATES

- ☐ Typecheck
- ☐ AI contract
- ☐ D.3
- ☐ Authentication
- ☐ API security
- ☐ Security
- ☐ Release
- ☐ Mobile offline sync
- ☐ End-to-end workflow
- ☐ Production smoke test

STATUS VALUES:

PASS

FAIL

NOT RUN

NOT APPLICABLE

Missing evidence:

NOT VERIFIED

=====

=====

19. DEPLOYMENT GATE

Before production advancement:

- ☐ Backup verified
- ☐ Tests pass
- ☐ Commit created
- ☐ Commit pushed
- ☐ Remote SHA verified
- ☐ Deployment created
- ☐ Deployment status verified
- ☐ Production URL verified
- ☐ HTTP response verified
- ☐ Authentication verified
- ☐ Core workflow verified
- ☐ AI provider status verified
- ☐ Database status verified
- ☐ Rollback path known

=====

=====

20. PRODUCTION GATE

Production is not verified until actual evidence exists.

VERIFY:

- ☐ URL
- ☐ Deployment
- ☐ HTTP status
- ☐ Login
- ☐ Workspace
- ☐ Lead

- ☐ Inspection
- ☐ Photo upload
- ☐ AI analysis
- ☐ Technician verification
- ☐ Measurement
- ☐ Estimate
- ☐ Report
- ☐ Mobile where applicable

=====

=====

21. PILOT GATE

A real pilot exercises:

- ☐ Real customer
- ☐ Real property
- ☐ Real inspection
- ☐ Real photographs
- ☐ Real technician
- ☐ Real measurements
- ☐ Real estimate
- ☐ Real report

FAILURE LOOP:

OBSERVE

- RECORD
- REPRODUCE
- FIX
- TEST
- COMMIT
- PUSH
- VERIFY
- DEPLOY
- VERIFY AGAIN

=====

=====

22. OWNER / TIME-BOX MODEL

Every active milestone requires:

OWNER:

TIME BOX:

START:

EXPECTED COMPLETION:

DEFINITION OF DONE:

REQUIRED EVIDENCE:

No milestone is complete merely because work began.

=====

=====

23. SINGLE ROOF/OS STATUS TEMPLATE

ROOF/OS — STATUS REPORT

DATE:

TIME:

REPORTING OWNER:

CURRENT ROADMAP PHASE:

CURRENT EXECUTION MILESTONE:

OVERALL STATUS:

[] GREEN — On track

[] YELLOW — Blocked / at risk

[] RED — Critical failure

[] BLUE — External dependency

[] NOT VERIFIED — Evidence missing

ONE-LINE STATUS:

CURRENT OBJECTIVE:

CURRENT REPO:

REPO URL:

BRANCH:

COMMIT:

COMMIT MESSAGE:

REMOTE VERIFIED: YES / NO

REMOTE SHA:

WORKTREE: CLEAN / DIRTY

UNCOMMITTED CHANGES:

BACKUP:

VERIFIED: YES / NO

BRANCH:

COMMIT:

REMOTE VERIFICATION:

RECOVERY TESTED: YES / NO

IF NO, WHY:

CURRENT EXECUTION MILESTONE:

OWNER:

START:

TIME BOX:

EXPECTED COMPLETION:

DEFINITION OF DONE:

WORK COMPLETED:

- ☐ Code
- ☐ Database
- ☐ Security
- ☐ Authentication
- ☐ AI
- ☐ Mobile
- ☐ Testing
- ☐ Deployment
- ☐ Documentation
- ☐ Other

DETAILS:

CONCRETE EVIDENCE:

TESTS:

TEST RESULTS:

COMMANDS / VERIFICATION:

PRODUCTION EVIDENCE:

DATABASE EVIDENCE:

SECURITY EVIDENCE:

REMOTE GIT EVIDENCE:

FAILURES / BLOCKERS:

1.

DESCRIPTION:

OWNER:

PRIORITY:

TIME BOX:

DEPENDENCY:

NEXT ACTION:

2.

DESCRIPTION:

OWNER:

PRIORITY:

TIME BOX:

DEPENDENCY:

NEXT ACTION:

TEST STATUS:

TYPECHECK:

AI CONTRACT:

D.3:

AUTH:

API SECURITY:

SECURITY:

RELEASE:

MOBILE:

END-TO-END:

PRODUCTION:

PRODUCTION:

URL:
DEPLOYMENT ID:
HTTP STATUS:
LOGIN:
CORE WORKFLOW:
AI PROVIDER:
DATABASE:
INCIDENT:

NEXT ACTION:

ONE NEXT ACTION:
OWNER:
TIME BOX:
EXPECTED EVIDENCE:

NEXT MILESTONE:

MILESTONE:
OWNER:
TIME BOX:
DEFINITION OF DONE:
REQUIRED EVIDENCE:

CHECKPOINT:

TESTED: YES / NO
COMMITTED: YES / NO
PUSHED: YES / NO
REMOTE VERIFIED: YES / NO
CLEAN: YES / NO
EVIDENCE RECORDED: YES / NO
BLOCKERS ASSIGNED: YES / NO
NEXT MILESTONE ASSIGNED: YES / NO
ADVANCE: YES / NO
REASON:

FINAL STATUS LINE:

STATUS:
EVIDENCE:
BLOCKER:
NEXT ACTION:
OWNER:
TIME BOX:
NEXT CHECKPOINT:

=====

=====

24. STATUS RULES

GREEN

On track with objective evidence.

YELLOW

Blocked or at risk.

RED

Critical failure.

BLUE

External dependency.

NOT VERIFIED

Required evidence does not exist.

NEVER CONVERT:

NOT VERIFIED

→

PASS

=====

=====

25. MANDATORY EXECUTION LOOP

BACKUP

→

TEST
→
COMMIT
→
PUSH
→
VERIFY
→
REPORT
→
ADVANCE

=====

=====

26. FINAL OPERATING LAW

THE ROADMAP TELLS US WHERE WE ARE GOING.

EXECUTION TELLS US WHAT WE ARE DOING NOW.

DO NOT CONFUSE THE TWO.

BUSINESS TARGET:

50 STATES

×

10 CUSTOMERS PER STATE

500 CUSTOMERS

500 CUSTOMERS

×

\$500/YEAR

\$250,000/YEAR ARR

NO LOST WORK.

NO BLIND CODING.

NO UNVERIFIED CLAIMS.

NO FAKE EVIDENCE.

NO BYPASSING AUTHORIZATION.

NO AI AUTHORITY OVER MEASUREMENTS.

NO AI AUTHORITY OVER PRICING.

NO AI AUTHORITY OVER INSURANCE DECISIONS.

NO UNCONTROLLED PRODUCTION CHANGES.

NO ADVANCING WITHOUT EVIDENCE.

SYSTEM MODEL:

AI OBSERVES.

ROOF/OS VALIDATES.

GEOMETRY CALCULATES.

TECHNICIAN VERIFIES.

ESTIMATOR PRICES.

D.2 PROTECTS THE MEANING OF AI OUTPUT.

D.3 PROTECTS THE PROCESS THAT PRODUCES AND STORES AI OUTPUT.

EVERY IMPORTANT CHANGE MUST BE RECOVERABLE.

EVERY IMPORTANT CLAIM MUST HAVE EVIDENCE.

EVERY MILESTONE MUST HAVE AN OWNER.

EVERY BLOCKER MUST HAVE A NEXT ACTION.

EVERY ADVANCE MUST BE EARNED.

=====

=====

END OF ROOF/OS MASTER PLAYBOOK